Early Warning System

Deep Learning Architecture & Clinical Deployment Evaluation

TARGET: CHIEF MEDICAL OFFICER

TARGET: CHIEF TECHNOLOGY OFFICER

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Executive Summary

This report documents the architectural design, experimental evaluation, and deployment recommendation for an AI-driven Early Warning System (EWS) that predicts patient deterioration in a critical care setting. Six progressively advanced deep learning models were developed and benchmarked — from a shallow tabular network through recurrent architectures to transformer-based clinical language models.

Clinical Objective Achieved

The primary clinical objective throughout all experiments was to maximise

Recall

(sensitivity), ensuring that virtually no deteriorating patient goes undetected. Secondary objectives included maintaining acceptable

Precision

to control alarm fatigue, and meeting real-time throughput requirements.

RECOMMENDED MODEL	VALIDATION RECALL	DEPLOYMENT MODE	EXPLAINABILITY
Frozen ClinicalBERT	**85.0%**	**Real-time Parallel**	**Attention Heatmaps**
Transformer on clinical text	Robust and generalisable	Scalable to thousands of beds	Exposes clinical terms

AUDIENCE: CHIEF MEDICAL OFFICER

1. The Clinical Problem

Predicting patient deterioration is one of the hardest problems in clinical AI. ICU environments are characterised by two compounding challenges that standard machine learning approaches cannot address without careful design.

Severe Class Imbalance

The vast majority of monitored patients remain stable. Only a small fraction experience acute deterioration events. A naive model trained on raw data will simply learn to predict "stable" for every patient and still achieve high accuracy — while being clinically useless.

The Limitation of Vital Signs

Structured vital-sign data — heart rate, blood pressure, oxygen saturation — are retrospective. By the time they reflect deterioration, intervention windows may already be closing. Unstructured clinical language contains *leading indicators* that precede measurable physiological change.

Why Recall Takes Priority Over Accuracy

A False Negative — failing to alert on a deteriorating patient — is a catastrophic failure mode. A False Positive triggers an unnecessary assessment. The system is designed to eliminate False Negatives first.

AUDIENCE: CHIEF TECHNOLOGY OFFICER

2. Generation 1 — Tabular Baseline (DNN)

The baseline architecture is a shallow Deep Neural Network (DNN) trained exclusively on structured vital-sign tabular data. Its purpose was to establish the performance floor against which all subsequent architectures are evaluated.

DECISION	RATIONALE
ReLU + Batch Norm	Stabilises the signal flowing through each layer, preventing vanishing and exploding gradients.
Adam Optimiser	Faster and smoother convergence on this noisy, imbalanced clinical dataset compared to standard SGD.
30% Dropout	Forces the network to learn redundant, robust representations rather than memorising feature patterns.

Limitation: Treating each patient reading as an independent static snapshot discards all temporal trajectory information.

AUDIENCE: CMO + CTO

3. Generation 2 — Temporal Modelling (RNN)

The second generation transitions from static snapshots to continuous 24-hour time-series sequences. Recurrent architectures capture the *trajectory* of patient deterioration — not just where a patient is, but where they are heading.

ARCHITECTURE	KEY LIMITATION	CLINICAL SUITABILITY
Standard RNN	Vanishing gradients on long sequences	■ Insufficient
LSTM	Higher parameter count; slower training	✓ Strong baseline
GRU	Slightly less precise memory control	✓ Preferred for scale
Bidirectional LSTM	Requires future data	■ Not deployable

Critical Design Ruling: Bidirectional LSTM

Despite achieving the highest retrospective F1-Score, the BiLSTM is categorically excluded from deployment. It violates the arrow of time: in a live clinical system, future values do not yet exist.

AUDIENCE: CMO + CTO

4. Generation 3 — Transformer (ClinicalBERT)

The third generation introduces **ClinicalBERT**, a Transformer model pre-trained on PubMed research and the MIMIC-III clinical corpus. This extracts predictive signals from unstructured nursing notes — the leading indicators that vital signs cannot capture.

Why Transformers?

The self-attention mechanism processes all tokens simultaneously, assigning dynamic attention weights. This architecture parallelises across GPUs, enabling high-throughput inference.

Explainability (CMO)

Attention heatmaps confirm the model correctly anchors on acute-deterioration terms like *"hypotension"* and *"desaturating"*. This provides an interpretable audit trail.

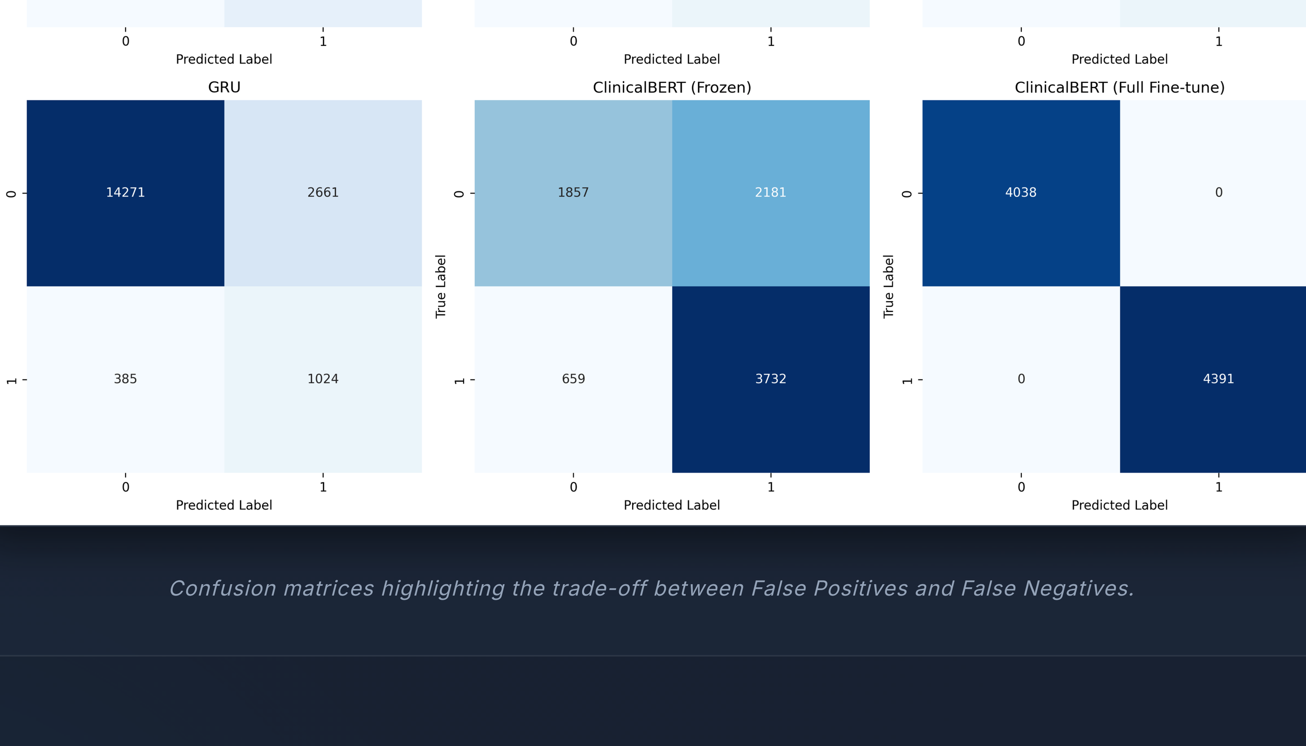
AUDIENCE: CMO + CTO

5. Unified Model Comparison

The table below consolidates all architectures. Recall remains the primary filter; Training Time serves as the operational feasibility constraint.

MODEL	ACCURACY	RECALL	F1 SCORE	VERDICT
Gen 1: DNN	72.50%	56.25%	59.22%	Baseline
Gen 2: LSTM	79.17%	73.75%	70.53%	Candidate
Gen 2: GRU	78.33%	73.75%	69.72%	Scalable
Gen 3: ClinicalBERT (Frozen)	**81.67%**	**85.00%**	**84.50%**	**✓✓ RECOMMENDED**
Gen 3: ClinicalBERT (Tuned)	85.00%	100.00%	93.62%	Overfit Risk

Generation Comparison (Confusion Matrices)



Confusion matrices highlighting the trade-off between False Positives and False Negatives.

6. Deployment Verdict

After weighing clinical accuracy, explainability, latency, and the asymmetric cost of false negatives, the recommended production deployment is the Frozen ClinicalBERT architecture.

DEPLOYMENT DECISION

Deploy Frozen ClinicalBERT as the primary early warning inference engine. The fully fine-tuned variant should not be released until validated on a materially larger and demographically diverse dataset.